Dynamical Flows and Probability Transport

A framework for algebraic solution and subsequent research

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Framework, core problems, and research

Constructing the framework

1. States, observables, and probability

2. Solving dynamics with evaluation and kernels

3. From flow geometry to probability transport

Using the framework

4. Operators, duality, and statistical evolution

5. Conservation and variational solution

6. Algebraic tools and subsequent research

The main question

How do algebraic structures turn dynamical problems into constructions that we can solve, compare, and approximate?

Appendix: core derivations

Manuscript § 1.2 and prospective connections to §§ 1.3–1.5.

States, observables, and probability laws

A common language for dynamics and statistics

$$x \in V, \qquad f \in \mathcal{A} = B_b(V), \qquad \mu \in \mathcal{P}(V).$$

The state specifies one configuration. An observable assigns a measured value to each state. A probability law describes an ensemble of configurations.

The pairing connects measurements to ensembles

$$\langle f, \mu \rangle = \int_V f(x) \mu(\,\mathrm{d}x).$$

The framework question

Given an evolution of states, construct compatible evolutions of observables and probability laws, then identify the structures they preserve. The examples below use linear maps, composition, and duality to carry out this construction.

Manuscript §§ 1.1–1.2, state–observable–probability framework.

The algebraic structures of the framework

Linear maps on states

$$V = \mathbb{R}^d, \qquad \mathrm{End}(V) \cong M_d(\mathbb{R}).$$

$$AB = A \circ B, \qquad I_V x = x.$$

Composition is associative and generally noncommutative.

Linear equations on trajectories

Kernels describe homogeneous solutions. A particular solution translates a kernel into an affine family.

The observable algebra

Bounded Borel functions form $\mathcal{A} = B_b(V)$, with pointwise operations:

$$(fg)(x) = f(x)g(x), \qquad 1(x) = 1.$$

This algebra is commutative and unital.

Functions and measures

The pairing $\langle f, \mu \rangle$ transfers an operator on observables to an evolution of laws.

Each structure supplies a specific operation for solving or interpreting the dynamics.

Manuscript §1.1 and §1.2.1. Algebra operations: Appendix A1.

From a velocity field to an ODE

A fluid element is carried by the local fluid velocity. Its position $x_t \in V$ evolves from an initial position x at time s according to

$$\begin{cases} \dot{x}_t = v_t(x_t), & t \in I, \\ x_s = x. \end{cases}$$

Here $I \subseteq \mathbb{R}$ is an open time interval, with $(s, x) \in I \times V$.

Velocity field

$$v : I \times V \longrightarrow V, \qquad (t, z) \longmapsto v_t(z),$$
$$v_t : V \longrightarrow V, \qquad z \longmapsto v_t(z).$$

At a fixed time t, v_t assigns a velocity to every position z.

Trajectory

$$x_\cdot : I \longrightarrow V, \qquad t \longmapsto x_t, \qquad x_\cdot \in C^1(I, V).$$

The solution traces one fluid element through time. Its tangent $\dot{x}_t$ equals the field evaluated at the current position, $v_t(x_t)$.

Manuscript §1.2.1, fluid-motion background and ODE initial-value problems.

One state flow induces compatible evolutions

State evolution

$$x_t = \Phi_{s,t}(x_s), \qquad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t}.$$

Pullback of observables and pushforward of laws

$$\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}, \qquad \mu_t = \mathcal{K}^*_{s,t} \mu_s := (\Phi_{s,t})_{\#} \mu_s.$$

$$\boxed{\langle \mathcal{K}_{s,t} f, \mu_s \rangle = \langle f, \mathcal{K}^*_{s,t} \mu_s \rangle.}$$

The algebra respects composition

$$\mathcal{K}_{s,r} \mathcal{K}_{r,t} = \mathcal{K}_{s,t}, \qquad \mathcal{K}^*_{r,t} \mathcal{K}^*_{s,r} = \mathcal{K}^*_{s,t}.$$

This compatibility lets us pass between trajectories, measurements, and distributions.

Manuscript §1.1, transformation duality, and §1.2.2, dynamical evolution.

An affine ODE as a linear operator equation

Given data

An open interval $I \subseteq \mathbb{R}$, with

$$A \in C(I, \mathrm{End}(V)),$$
$$B \in C(I, V).$$

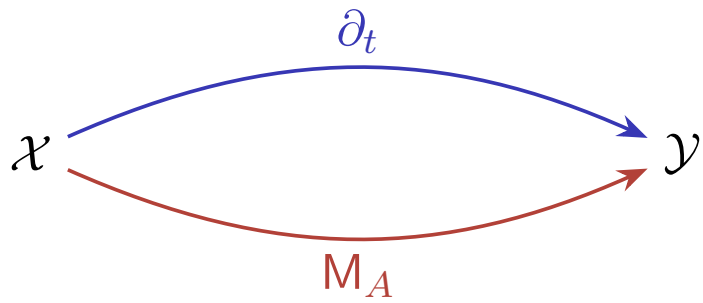
For $(s, x) \in I \times V$, solve

$$\begin{cases} \dot{x}_t = A_t x_t + B_t, & t \in I, \\ x_s = x. \end{cases}$$

Continuous coefficients give a unique trajectory on all of I.

$$\mathcal{X} = C^1(I, V), \qquad \mathcal{Y} = C(I, V),$$

Trajectory formulation



$$\mathsf{D}_A = \partial_t - \mathsf{M}_A$$

$$\mathsf{D}_A \gamma = B, \qquad E_s \gamma = x.$$

The unknown is a complete trajectory.

$$(\mathsf{M}_A \gamma)_t = A_t \gamma_t.$$

Assessment 9.1; Manuscript Example 1.10. Existence and uniqueness: Appendix A2.

Homogeneous evolution by evaluation

The homogeneous kernel

$$\mathcal{H} = \ker \mathsf{D}_A.$$

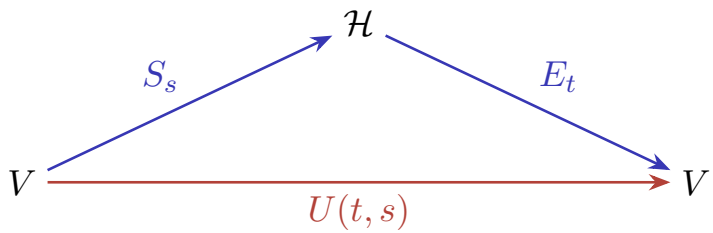
Evaluation identifies a trajectory with its state:

$$E_s : \mathcal{H} \to V, \qquad E_s(\eta) = \eta_s.$$

Existence and uniqueness make E_s a linear isomorphism.

$$S_s = E_s^{-1}, \qquad U(t,s) = E_t S_s.$$

Recover, then evaluate



Write $U_t = U(t,s)$. Then

$$\dot{U}_t = A_t U_t,$$
$$U_s = I_V.$$

Manuscript Example 1.10. Evaluation isomorphism and composition: Appendix A3.

The affine solution and its flow

Homogeneous and forced response

$$x_t = U(t,s)x + a_{s,t},$$

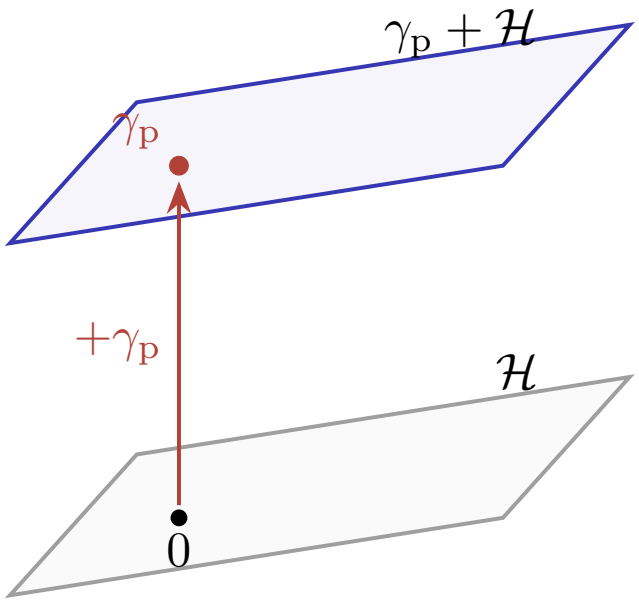
$$a_{s,t} = \int_s^t U(t,r)B_r \, \mathrm{d}r.$$

The initial value selects one trajectory from an affine family.

Initial-to-terminal state map

$$\Phi_{s,t}(x) = U(t,s)x + a_{s,t}.$$

For zero forcing, the map reduces to $U(t,s)x$.



Linear and affine subsets of $\mathcal{X}$

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H}.$$

Manuscript Example 1.10. Affine family: A4; derivation: A5; verification: A6.

Evaluation also constructs nonlinear flows

Allow a nonlinear velocity field

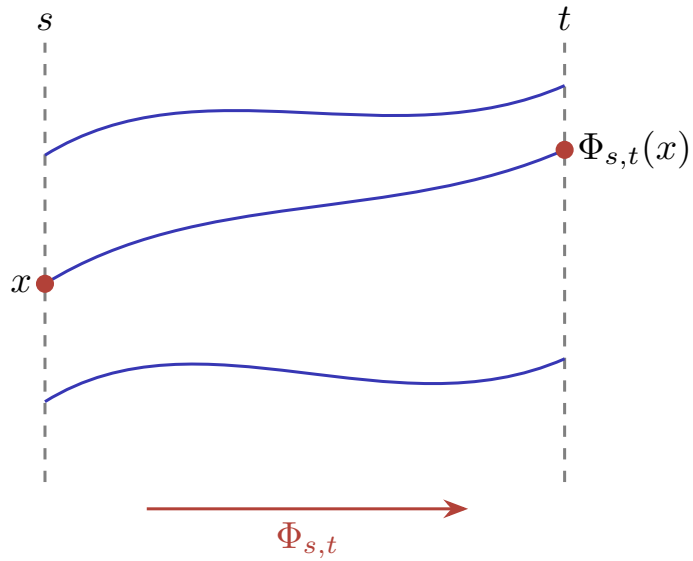
$$\dot{x}_t = v_t(x_t), \qquad x_s = x, \qquad \Phi_{s,t}(x) = x_t.$$

Assume continuous, locally Lipschitz dynamics with a unique trajectory on the same interval I for every initial pair.

The algebraic construction remains

$$\Phi_{s,t} = E_t \circ E_s^{-1}.$$

Evaluation is a bijection of sets; linearity of the state map is unnecessary.



Assessment 9.2; Manuscript Theorem 1.12. General evaluation construction: Appendix A7.

Flow laws by cancellation of evaluation maps

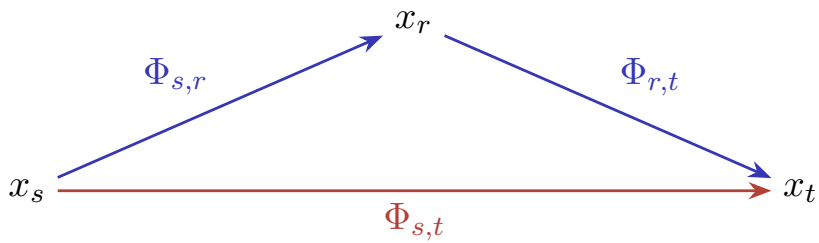
On the set $\mathcal{S}_v$ of complete solution trajectories, existence and uniqueness make each evaluation $E_s : \mathcal{S}_v \to V$ a bijection.

Composition follows from an inverse identity

$$\begin{aligned}\Phi_{r,t} \circ \Phi_{s,r} &= (E_t \circ E_r^{-1}) \circ (E_r \circ E_s^{-1}) \\ &= E_t \circ E_s^{-1} = \Phi_{s,t}.\end{aligned}$$

Identity and inverse

$$\Phi_{s,s} = I_V, \qquad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$



The identities hold on a common interval for all initial data.

The flow identities use composition and invertibility of evaluation. Linearity of the state map is unnecessary.

Manuscript §1.2.1; Assessment 9.2. Evaluation construction: Appendix A7.

The derivative of a flow

Perturbing the initial state

For a C^1 state flow and a small displacement δx,

$$\Phi_{s,t}(x+\delta x) = \Phi_{s,t}(x) + M_{s,t}(x)\delta x + o(\|\delta x\|), \qquad M_{s,t}(x) = D_x\Phi_{s,t}(x).$$

Variational equation

Differentiating $\partial_t \Phi_{s,t}(x) = v_t(\Phi_{s,t}(x))$ in x gives

$$\begin{cases} \partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x))M_{s,t}(x), \\ M_{s,s}(x) = I_d. \end{cases}$$

The nonlinear trajectory determines the coefficients of a linear matrix ODE.

Composition of derivatives

$$M_{s,t}(x) = M_{r,t}(\Phi_{s,r}(x))M_{s,r}(x).$$

The intermediate matrix is evaluated at the transported state.

Manuscript § 1.2.1, variational equation; Assessment 9.3–4.

Time ordering in matrix evolution

Fix a trajectory and set $A_t = Dv_t(\Phi_{s,t}(x))$. Then

$$\dot{M}_t = A_t M_t, \qquad M_s = I_d.$$

Successive substitution, for $t \geq s$

$$M_t = I_d + \int_s^t A_{r_1}\,\mathrm{d}r_1 + \int_s^t \int_s^{r_1} A_{r_1} A_{r_2}\,\mathrm{d}r_2\,\mathrm{d}r_1 + \cdots .$$

Later times appear on the left in each operator product.

General coefficients

$$M_t = \mathcal{T} \exp\left(\int_s^t A_r\,\mathrm{d}r \right).$$

The order matters when $A_r A_u \neq A_u A_r$.

Pairwise commuting coefficients

If $A_r A_u = A_u A_r$ for every pair,

$$M_t = \exp\left(\int_s^t A_r\,\mathrm{d}r \right).$$

Manuscript §1.2.1, Peano–Baker series, and Appendix A.5.

Jacobian determinant and local volume

Liouville–Jacobi formula

For a C^1 diffeomorphic flow, define $J_{s,t}(x) = \det M_{s,t}(x)$. Then

$$\partial_t J_{s,t}(x) = \nabla \cdot v_t(\Phi_{s,t}(x)) J_{s,t}(x), \qquad J_{s,s}(x) = 1,$$

and therefore

$$\boxed{J_{s,t}(x) = \exp\left(\int_s^t \nabla \cdot v_r(\Phi_{s,r}(x))\, \mathrm{d}r \right) > 0.}$$

Geometric meaning

- $J_{s,t}$ is the local volume ratio of a transported material element.
- Positive divergence gives instantaneous expansion. Negative divergence gives contraction.
- A divergence-free field gives $J_{s,t} = 1$, allowing shear and deformation.

Manuscript §1.2.1; Assessment 9.3–4. Jacobi's formula: Appendix A8.

Volume change in the nonlinear example

For $\dot{x} = 0.2x$, $\dot{y} = 0.1y + 0.3x^2$, set $\tau = t - s$. The exact flow is

$$\Phi_{s,t}(x, y) = \begin{pmatrix} e^{0.2\tau} x \\ e^{0.1\tau} y + x^2 (e^{0.4\tau} - e^{0.1\tau}) \end{pmatrix}.$$

Differentiate the explicit solution

$$M_{s,t}(x, y) = \begin{pmatrix} e^{0.2\tau} & 0 \\ 2x(e^{0.4\tau} - e^{0.1\tau}) & e^{0.1\tau} \end{pmatrix}, \qquad J_{s,t} = e^{0.3\tau}.$$

Check against divergence

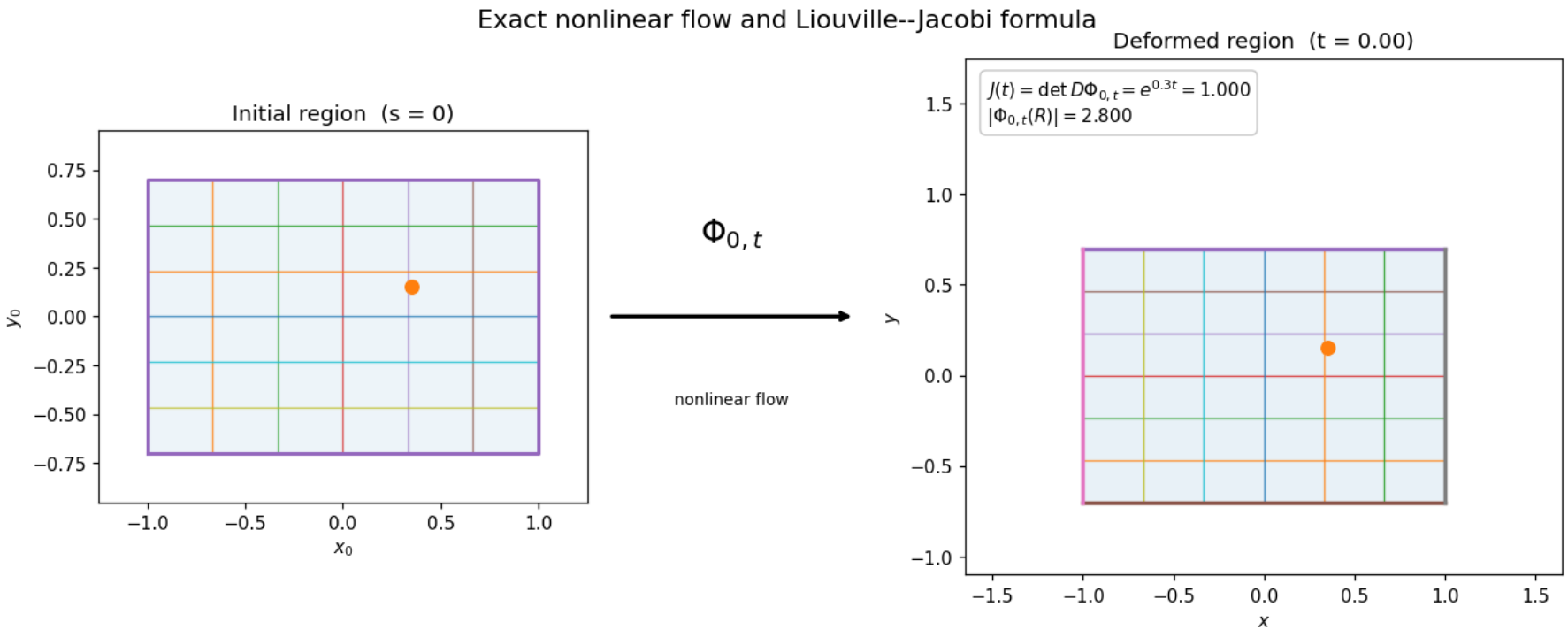
$$\nabla \cdot v = 0.2 + 0.1 = 0.3, \qquad \exp\left(\int_s^t 0.3 \, \mathrm{d}r \right) = e^{0.3\tau}.$$

The quadratic term changes shape while the area factor is independent of the initial state. The explicit solution guarantees completeness even though the field is not globally Lipschitz.

Manuscript §1.2.1, nonlinear volume-deformation example. Return to the animation.

A nonlinear flow in motion

$$R = [-1, 1] \times [-0.7, 0.7], \qquad 0 \le t \le 1.5.$$



The same flow moves every grid point. The quadratic coupling bends the rectangle.

Manuscript Example 1.15 and its accompanying 61-frame animation.

Density transport by a diffeomorphism

Assume $\mu_0(\,\mathrm{d}x) = \rho_0(x)\,\mathrm{d}x$ and a C^1 diffeomorphic flow.

Change of variables

$$\rho_t(y) = \rho_0(\Phi_{0,t}^{-1}(y))\left|\det D\Phi_{0,t}^{-1}(y)\right|.$$

Along $x_t = \Phi_{0,t}(x_0)$, the formula becomes

$$\boxed{\rho_t(x_t)J_{0,t}(x_0) = \rho_0(x_0).}$$

The divergence controls density along a trajectory

$$\rho_t(x_t) = \rho_0(x_0)\exp\left(-\int_0^t \nabla\cdot v_r(x_r)\,\mathrm{d}r\right).$$

Expansion lowers density within a material element. Its probability mass is conserved:

$$\mu_t(\Phi_{0,t}(A)) = \mu_0(A).$$

These density identities hold almost everywhere for $\rho_0 \in L^1$.

Manuscript §1.2.2; Assessment 9.7–8. Change of variables: Appendix A10.

The classical continuity equation

For the transport representative of the density, differentiation along almost every trajectory gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_t(x_t) = -\rho_t(x_t)\nabla \cdot v_t(x_t).$$

If $\rho(t,x)$ is also C^1, the chain rule yields

$$\partial_t \rho_t + v_t \cdot \nabla \rho_t = -\rho_t \nabla \cdot v_t, \qquad \boxed{\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0.}$$

Generator and adjoint

For $\mathcal{L}_t f = v_t \cdot \nabla f$ and compactly supported smooth f,

$$\int (\mathcal{L}_t f)\rho \,\mathrm{d}x = -\int f \nabla \cdot (v_t \rho)\,\mathrm{d}x, \qquad \mathcal{L}_t^* \rho = -\nabla \cdot (v_t \rho).$$

Thus $\partial_t \rho_t = \mathcal{L}_t^* \rho_t$.

Measure formulation

Without a density, use $\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0$ in distributions, under local flux integrability.

Manuscript §1.2.2. Weak formulation: Appendix A11.

Evolution on the observable algebra

Pull back the observable

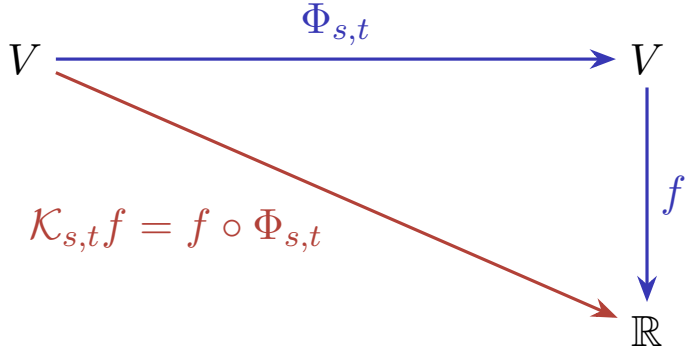
On the observable algebra $\mathcal{A} = B_b(V)$, define

$$\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}.$$

The state map may be nonlinear, while its induced action on functions is linear.

$$\mathcal{K}_{s,t}(fg) = (\mathcal{K}_{s,t} f)(\mathcal{K}_{s,t} g),$$
$$\mathcal{K}_{s,t} 1 = 1.$$

$$\mathcal{K}_{s,r}\mathcal{K}_{r,t} = \mathcal{K}_{s,t}.$$



Manuscript §1.1, transformation and evolution operators. State maps are Borel measurable.

Observable evolution and probability evolution

For a bounded Borel observable f, the pushforward identity gives

$$\boxed{\langle f, \mu_t \rangle = \int f(\Phi_{s,t}(x))\mu_s(\,\mathrm{d}x) = \langle \mathcal{K}_{s,t} f, \mu_s \rangle.}$$

Schrödinger picture

Keep the observable fixed and evolve the law:

$$\mu_t = (\Phi_{s,t})_{\#}\mu_s.$$

Integrate f over the current distribution.

Heisenberg picture

Keep the initial law fixed and pull back the observable:

$$\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}.$$

Integrate the evolved observable over μ_s.

Both descriptions give the same expectation. These identities also hold for unbounded observables when the relevant integrals are finite.

Manuscript §1.2.2, observable–measure duality and dynamical pictures.

Infinitesimal evolution of observables

The deterministic generator on smooth test functions

$$\mathcal{L}_t f = v_t \cdot \nabla f = \lim_{h \downarrow 0} \frac{\mathcal{K}_{t,t+h} f - f}{h}.$$

The limit is understood on an appropriate smooth domain and in the chosen topology.

Differentiating the two time arguments

For sufficiently regular f and a smooth invertible flow,

$$\partial_t \mathcal{K}_{s,t} f = \mathcal{K}_{s,t} \mathcal{L}_t f, \qquad \partial_s \mathcal{K}_{s,t} f = -\mathcal{L}_s (\mathcal{K}_{s,t} f).$$

The terminal-time derivative acts on f before pullback. The initial-time derivative acts on the already pulled-back function.

Evolution of ensemble averages

Under integrable domination along trajectories,

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle f, \mu_t \rangle = \langle \mathcal{L}_t f, \mu_t \rangle.$$

Manuscript §1.2.2, generators and observable evolution. Time derivatives: Appendix A9.

Markov evolution extends the operator framework

Average over a transition law

For a transition kernel $\mathsf{K}_{s,t}(x, \mathrm{d}y)$, define

$$(\mathsf{P}_{s,t} f)(x) = \int f(y) \mathsf{K}_{s,t}(x, \mathrm{d}y).$$

The same observable–measure pairing gives

$$\langle f, \mu_t \rangle = \langle \mathsf{P}_{s,t} f, \mu_s \rangle, \qquad \mathsf{P}_{s,r} \mathsf{P}_{r,t} = \mathsf{P}_{s,t}.$$

Which algebraic properties survive?

Markov operators are linear, positive, and preserve the constant 1. They generally do not preserve products of observables.

Deterministic dynamics

$$\mathsf{K}_{s,t}(x, \mathrm{d}y) = \delta_{\Phi_{s,t}(x)}(\mathrm{d}y) \quad \Longrightarrow \quad \mathsf{P}_{s,t} = \mathcal{K}_{s,t}.$$

In this special case, pullback also satisfies $\mathcal{K}(fg) = (\mathcal{K}f)(\mathcal{K}g)$.

Manuscript §1.2.2, Markov kernels and deterministic evolution as a special case.

Differential entropy under deterministic transport

For a smooth positive density, define

$$S(\rho_t) = -\int \rho_t \log \rho_t \, \mathrm{d}x, \qquad s_t^\rho = \nabla \log \rho_t.$$

Assume a classical continuity equation, finite integrals, justified differentiation, and vanishing boundary terms.

Entropy rate

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} S(\rho_t) = \mathbb{E}_{\rho_t}[\nabla \cdot v_t] = -\mathbb{E}_{\rho_t}[v_t \cdot s_t^\rho].}$$

- Positive mean divergence increases differential entropy.
- Negative mean divergence decreases it.
- Hamiltonian transport preserves entropy under these assumptions.

For the triangular example, $S(\rho_t) = S(\rho_0) + 0.3t$ when the entropies are finite.

Manuscript §1.2.4; Assessment 9.10. Entropy derivation: Appendix A13.

Hamiltonian dynamics in phase space

For unit mass, let $z = (q, p) \in \mathbb{R}^{2n}$ and

$$H(q,p) = \tfrac{1}{2}\|p\|^2 + \mathcal{V}(q), \qquad \mathcal{V} \in C^2(\mathbb{R}^n).$$

Hamilton's equations

$$\dot{q}_t = p_t, \qquad \dot{p}_t = -\nabla \mathcal{V}(q_t), \qquad \dot{z}_t = \mathbb{J}_n \nabla H(z_t),$$

where

$$\mathbb{J}_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \qquad \mathbb{J}_n^\top = -\mathbb{J}_n, \qquad \mathbb{J}_n^2 = -I_{2n}.$$

Canonical symplectic form

$$\omega(u,w) = u^\top \mathbb{J}_n w = u_q^\top w_p - u_p^\top w_q.$$

We study the flow and its derivative on their interval of existence.

Manuscript §1.2.3, Hamiltonian phase-space dynamics; Assessment Project 10.

Energy, symplecticity, and phase-space volume

Let ϕ_t be the Hamiltonian flow and $M_t = D\phi_t$.

Energy conservation

$$\frac{\mathrm{d}}{\mathrm{d}t} H(z_t) = \nabla H(z_t)^\top \mathbb{J}_n \nabla H(z_t) = 0.$$

Preservation of the symplectic form

With $A_t = \mathbb{J}_n \nabla^2 H(z_t)$, symmetry of the Hessian gives

$$A_t^\top \mathbb{J}_n + \mathbb{J}_n A_t = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(M_t^\top \mathbb{J}_n M_t) = 0.$$

Since $M_0 = I_{2n}$, we obtain $M_t^\top \mathbb{J}_n M_t = \mathbb{J}_n$.

Liouville's theorem

$$\nabla \cdot v = \operatorname{tr}(\mathbb{J}_n \nabla^2 H) = 0, \qquad \det M_t = 1.$$

The flow preserves phase-space volume and transports density without changing its value along trajectories.

Manuscript §1.2.3; Assessment 10.2 and 10.4. Symplectic calculation: Appendix A12.

A variational problem solved by a change of measure

For a reference law ν and bounded measurable energy U, consider

$$\mathcal{F}_U(\mu) = \mathrm{KL}(\mu\|\nu) + \int U \,\mathrm{d}\mu.$$

Here $\mathrm{KL}(\mu\|\nu) = \int \log(\mathrm{d}\mu/\mathrm{d}\nu)\,\mathrm{d}\mu$ for $\mu \ll \nu$, and $+\infty$ otherwise. It is nonnegative, with equality exactly when $\mu = \nu$.

Tilt the reference probability law

$$Z_U = \int e^{-U}\,\mathrm{d}\nu, \qquad \nu_U = Z_U^{-1} e^{-U}\nu.$$

The Radon–Nikodym chain rule rewrites the entire objective:

$$\boxed{\mathcal{F}_U(\mu) = \mathrm{KL}(\mu\|\nu_U) - \log Z_U.}$$

$$\mu_* = \nu_U, \qquad \min_{\mu} \mathcal{F}_U(\mu) = -\log Z_U.$$

Changing the reference law turns minimization into the equality case of a nonnegative quantity.

Manuscript §1.2.4; Assessment 11.1 and 11.5. Change-of-measure calculation: Appendix A14.

Algebraic tools supplied by the framework

A trajectory equation becomes a kernel and a translate

$$\mathsf{D}_A \gamma = B \quad \Longrightarrow \quad \mathcal{S}_B = \gamma_{\mathrm{p}} + \ker \mathsf{D}_A.$$

Evaluation turns uniqueness into composition

$$\Phi_{s,t} = E_t E_s^{-1}, \qquad \Phi_{r,t}\Phi_{s,r} = E_t E_r^{-1} E_r E_s^{-1}.$$

Pullback turns state motion into a linear action

$$\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}, \qquad \langle \mathcal{K}_{s,t} f, \mu \rangle = \langle f, (\Phi_{s,t})_{\#} \mu \rangle.$$

Trace, determinant, and tilting simplify derived problems

The trace reduces local volume evolution to a scalar ODE. Gibbs tilting reduces a measure optimization problem to KL nonnegativity.

Synthesis of the constructions in manuscript §§ 1.2.1–1.2.4.

Research direction: projection to collective variables

Choose $\xi : V \to \mathcal{Z}$ and write $Z_t = \xi(X_t)$, $\nu_t = \xi_{\#}\mu_t$.

Pullback embeds reduced observables

$$C_\xi g = g \circ \xi.$$

On L^2, conditional expectation is an orthogonal projection onto functions of Z_t.

A candidate reduced generator

For suitable test functions and integrable terms, define

$$(\overline{\mathcal{L}}_t g)(z) = \mathbb{E}[(\mathcal{L}_t C_\xi g)(X_t) \mid Z_t = z].$$

This conditional projection matches the instantaneous evolution of reduced expectations.

Research question

Determine when it defines a well-posed reduced process, and measure the information lost in temporal correlations. Matching one-time laws alone does not establish dynamical closure.

Research direction based on manuscript §1.3, collective variables and conditional effective dynamics.

Research direction: conditional learning of flows

For a differentiable interpolation $I_t = \alpha_t X_0 + \lambda_t X_1$, seek a velocity that transports its one-time laws.

A projection supplies the population target

If $\dot{I}_t$ is square integrable, then

$$u_t^* = \operatorname*{arg\,min}_{u \in L^2(\mathrm{Law}(I_t))} \mathbb{E}\|u(I_t) - \dot{I}_t\|^2,$$

$$u_t^*(x) = \mathbb{E}[\dot{I}_t \mid I_t = x].$$

The conditional projection defines the target almost everywhere.

Proposed extension

Fit a velocity or Hamiltonian in a class encoding the required structure. Compare induced operators and transported laws with the analytic baselines.

What must be checked

Separate representation, sampling, optimization, and time-discretization errors. The population identity does not guarantee exact recovery by a finite model.

Research direction: manuscript § 1.4, conditional learning and structural approximation.

A research programme built on the framework

Mathematical foundation

Use the same state–observable–law pairing to formulate exact evolution, reduction, learning, and numerical approximation.

Next validation tasks

1. Use affine flows and the coupled OU benchmark to compare exact and reduced generators.
2. Test both transported one-time laws and two-time correlations when assessing a learned reduction.
3. Check the relevant invariants under discretization: probability mass and positivity, or Hamiltonian symplecticity.

The organizing question

Which algebraic identities survive each approximation, and what error appears when they fail?

The proposed extensions use the exact constructions as references for measuring the errors introduced by reduction and learning.

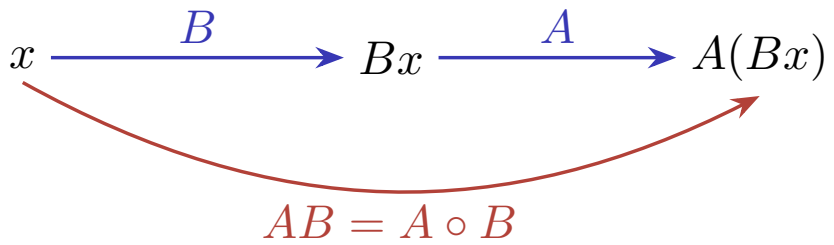
Research programme based on manuscript §§ 1.3–1.5 and the existing benchmark design.

Appendix A1: algebra operations

The algebra $\mathrm{End}(V)$

$$(A+B)x = Ax + Bx,$$
$$(aA)x = a(Ax),$$
$$(AB)x = A(Bx).$$

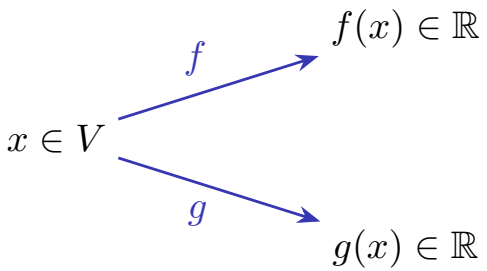
Composition is associative, with unit I_V; AB and BA can differ when $d \geq 2$.



The observable algebra $\mathcal{A} = B_b(V)$

$$(af + bg)(x) = af(x) + bg(x),$$
$$(fg)(x) = f(x)g(x),$$
$$1(x) = 1.$$

Bounded Borel functions form a commutative unital algebra.



Multiply values at the same state:
$(fg)(x) = f(x)g(x)$

Manuscript §1.1. Return to state space and algebra.

Appendix A2: existence and uniqueness

Bounds on a compact time window

For $v_t(y) = A_t y + B_t$ and compact $J \subset I$, let

$$L_J = \sup_{t \in J} \|A_t\|, \qquad C_J = \sup_{t \in J} \|B_t\|.$$

Both constants are finite, and

$$\|v_t(y) - v_t(z)\| \le L_J \|y - z\|,$$
$$\|v_t(y)\| \le C_J + L_J \|y\|.$$

Continuation argument

1. Local Lipschitz continuity gives a unique local solution.
2. Linear growth prevents finite-time escape on J.
3. Uniqueness joins the solutions on overlapping windows.

The trajectory is therefore uniquely defined on all of I.

Manuscript Example 1.10 and Appendix A.1. Return to the initial-value problem.

Appendix A3: evaluation and evolution

Homogeneous trajectories

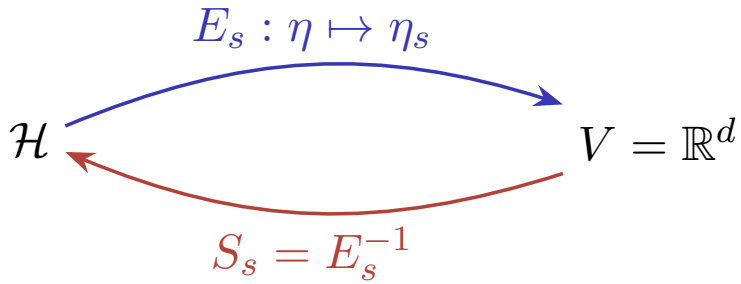
$$\mathcal{H} = \{\eta \in C^1(I, V) : \dot{\eta}_t = A_t \eta_t\}.$$

Evaluation $E_s(\eta) = \eta_s$ is linear.

- Existence makes it onto.
- Uniqueness makes it one-to-one.

$S_s = E_s^{-1}, \qquad \mathcal{H} \cong V, \qquad \dim \mathcal{H} = d.$

Inverse evaluation



$$E_s S_s = I_V, \qquad S_s E_s = I_{\mathcal{H}}.$$

Composition of the homogeneous evolution

$$U(t, r)U(r, s) = E_t S_r E_r S_s = E_t S_s = U(t, s).$$

Manuscript Example 1.10. Return to homogeneous evolution.

Appendix A4: the affine solution family

A particular solution shifts the kernel

Suppose $\mathsf{D}_A \gamma_{\mathrm{p}} = B$. Then

$$\begin{aligned} \mathsf{D}_A \gamma = B &\iff \mathsf{D}_A(\gamma - \gamma_{\mathrm{p}}) = 0 \\ &\iff \gamma - \gamma_{\mathrm{p}} \in \mathcal{H}. \end{aligned}$$

Hence

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H}.$$

The initial condition selects one path

Write $\gamma = \gamma_{\mathrm{p}} + \eta$. Then

$$E_s \eta = x - \gamma_{\mathrm{p}}(s).$$

Since E_s is an isomorphism,

$$\gamma = \gamma_{\mathrm{p}} + S_s\big(x - \gamma_{\mathrm{p}}(s)\big).$$

Taking $\gamma_{\mathrm{p}}(s) = 0$ gives

$$\gamma = \gamma_{\mathrm{p}} + S_s x.$$

Manuscript Example 1.10. Return to the affine solution.

Appendix A5: variation of constants

1. Vary the homogeneous coefficient

$$\gamma_{\mathrm{p}}(t) = U(t,s)c(t), \qquad c(s) = 0.$$

2. Apply the trajectory operator

$$\begin{aligned} \dot{\gamma}_{\mathrm{p}} - A_t\gamma_{\mathrm{p}} &= U(t,s)\dot{c}(t), \\ \dot{c}(t) &= U(s,t)B_t. \end{aligned}$$

3. Integrate and compose

$$c(t) = \int_s^t U(s,r)B_r \,\mathrm{d}r.$$

Composition gives $U(t,s)U(s,r) = U(t,r)$, hence

$$\gamma_{\mathrm{p}}(t) = \int_s^t U(t,r)B_r \,\mathrm{d}r.$$

$$x_t = U(t,s)x + \int_s^t U(t,r)B_r \,\mathrm{d}r.$$

Manuscript Example 1.10. Return to the solution formula.

Appendix A6: verification of the solution

Initial condition

$$x_s = U(s,s)x + \int_s^s U(s,r)B_r \,\mathrm{d}r = x.$$

Differentiation

Since $\partial_t U(t,r) = A_t U(t,r)$, Leibniz's rule gives

$$\begin{aligned}
\dot{x}_t &= A_t U(t,s)x + U(t,t)B_t + \int_s^t A_t U(t,r)B_r \,\mathrm{d}r \\
&= A_t \left(U(t,s)x + \int_s^t U(t,r)B_r \,\mathrm{d}r \right) + B_t \\
&= A_t x_t + B_t.
\end{aligned}$$

Uniqueness

The difference of two solutions lies in $\mathcal{H}$ and has zero value at s. Injectivity of E_s makes that difference the zero trajectory.

Manuscript Example 1.10. Return to the solution and its flow map.

Appendix A7: general evaluation and composition

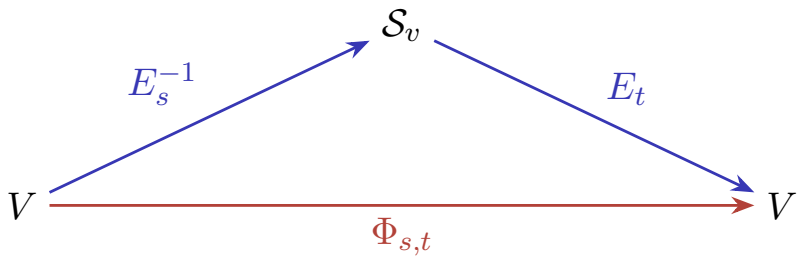
Evaluation is a bijection

$$\mathcal{S}_v = \{\gamma \in C^1(I, V) : \dot{\gamma}_t = v_t(\gamma_t)\}.$$

On the solution set, $E_s(\gamma) = \gamma_s$.
Existence gives surjectivity; uniqueness gives injectivity.

$$\Phi_{s,t} = E_t \circ E_s^{-1}.$$

Recover, then evaluate



Cancellation at an intermediate time

$$\begin{aligned}\Phi_{r,t} \circ \Phi_{s,r} &= (E_t \circ E_r^{-1}) \circ (E_r \circ E_s^{-1}) \\ &= E_t \circ E_s^{-1} = \Phi_{s,t}.\end{aligned}$$

$$\Phi_{s,s} = I_V, \qquad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$

Manuscript Appendix A.2. Return to general flows; return to the flow laws.

Appendix A8: the determinant equation

Invertibility from the inverse flow

$$D\Phi_{t,s}(\Phi_{s,t}(x))M_{s,t}(x) = I_d.$$

Thus $M = M_{s,t}(x)$ is invertible. Set $A_t = Dv_t(\Phi_{s,t}(x))$.

Jacobi's formula and cyclic invariance

$$\begin{aligned}\partial_t \det M &= (\det M)\,\mathrm{tr}(M^{-1}\partial_t M)\\ &= (\det M)\,\mathrm{tr}(M^{-1}A_t M)\\ &= (\det M)\,\mathrm{tr}(A_t).\end{aligned}$$

Since $\mathrm{tr}(Dv_t) = \nabla \cdot v_t$ and $\det M_{s,s} = 1$,

$$J_{s,t}(x) = \exp\left(\int_s^t \nabla \cdot v_r(\Phi_{s,r}(x))\,\mathrm{d}r\right).$$

The determinant equation is scalar even when the matrices at different times fail to commute.

Manuscript §1.2.1, variational matrices. Return to local volume.

Appendix A9: derivatives of the Koopman operator

Terminal time

For a smooth observable, the chain rule gives

$$\partial_t[f(\Phi_{s,t}(x))] = \nabla f(\Phi_{s,t}(x)) \cdot v_t(\Phi_{s,t}(x)) = (\mathcal{K}_{s,t}\mathcal{L}_t f)(x).$$

Initial time

The composition law yields

$$\Phi_{s,t} = \Phi_{s+h,t} \circ \Phi_{s,s+h}.$$

Differentiate at $h = 0$ to obtain

$$\partial_s \Phi_{s,t}(x) = -D_x \Phi_{s,t}(x) v_s(x).$$

Therefore,

$$\partial_s(\mathcal{K}_{s,t} f)(x) = -v_s(x) \cdot \nabla_x[f(\Phi_{s,t}(x))] = -\mathcal{L}_s(\mathcal{K}_{s,t} f)(x).$$

Manuscript § 1.2.2, generator identities. Return to infinitesimal evolution.

Appendix A10: density change of variables

For a Borel set A and a C^1 diffeomorphism $\Phi = \Phi_{0,t}$,

$$\begin{aligned}\mu_t(A) &= \mu_0(\Phi^{-1}(A)) \\ &= \int_{\Phi^{-1}(A)} \rho_0(x_0)\,\mathrm{d}x_0 \\ &= \int_A \rho_0(\Phi^{-1}(y))|\det D\Phi^{-1}(y)|\,\mathrm{d}y.\end{aligned}$$

Thus $\rho_t(y) = \rho_0(\Phi^{-1}(y))|\det D\Phi^{-1}(y)|$ almost everywhere.

The inverse Jacobian

$$D\Phi^{-1}(\Phi(x_0))D\Phi(x_0) = I_d, \qquad |\det D\Phi^{-1}(\Phi(x_0))| = J_{0,t}(x_0)^{-1}.$$

Substitution gives $\rho_t(\Phi(x_0))J_{0,t}(x_0) = \rho_0(x_0)$. Only integrability of ρ_0 is needed for this formula. A pointwise classical continuity equation requires the additional C^1 regularity of $\rho(t,x)$.

Manuscript §1.2.2, Assessment 9.7–8. Return to density transport.

Appendix A11: the measure-valued weak equation

For $\psi \in C_c^\infty((0,T)\times\mathbb{R}^d)$, define

$$F(t) = \int \psi(t,x)\mu_t(\,\mathrm{d}x) = \int \psi(t,\Phi_{0,t}(x_0))\mu_0(\,\mathrm{d}x_0).$$

The chain rule along trajectories and local flux integrability give

$$F'(t) = \int [\partial_t\psi + v_t\cdot\nabla\psi](t,x)\mu_t(\,\mathrm{d}x)$$

for almost every t. Integrating in time,

$$\int_0^T \int [\partial_t\psi + v_t\cdot\nabla\psi]\,\mathrm{d}\mu_t\,\mathrm{d}t = F(T) - F(0) = 0.$$

The endpoint terms vanish because ψ is supported away from $t = 0, T$.

Distributional interpretation

$$\langle \partial_t\mu + \nabla\cdot(v\mu), \psi\rangle = -\int_0^T \int [\partial_t\psi + v_t\cdot\nabla\psi]\,\mathrm{d}\mu_t\,\mathrm{d}t = 0.$$

Manuscript §1.2.2, weak continuity. Return to probability transport without a density.

Appendix A12: symplecticity of the exact flow

Write $Q_t = \nabla^2 H(z_t) = Q_t^\top$ and $\dot{M}_t = \mathbb{J}_n Q_t M_t$. Since $\mathbb{J}_n^\top = -\mathbb{J}_n$ and $\mathbb{J}_n^2 = -I$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(M_t^\top \mathbb{J}_n M_t) &= M_t^\top [(\mathbb{J}_n Q_t)^\top \mathbb{J}_n + \mathbb{J}_n (\mathbb{J}_n Q_t)] M_t \\ &= M_t^\top [Q_t(-\mathbb{J}_n)\mathbb{J}_n + \mathbb{J}_n^2 Q_t] M_t \\ &= M_t^\top (Q_t - Q_t) M_t = 0.\end{aligned}$$

The initial condition gives $M_t^\top \mathbb{J}_n M_t = \mathbb{J}_n$.

Volume preservation

Taking determinants gives $(\det M_t)^2 = 1$. Continuity in time and $\det M_0 = 1$ select $\det M_t = 1$. Alternatively, $\operatorname{tr}(\mathbb{J}_n Q_t) = 0$ because one factor is

skew-symmetric and the other symmetric. The Jacobian formula gives the same result.

Manuscript §1.2.3, Assessment 10.4. Return to geometric conservation.

Appendix A13: the entropy rate

Under the smoothness, finiteness, and boundary assumptions of the entropy theorem,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} S(\rho_t) &= -\int (1+\log \rho_t)\partial_t \rho_t \,\mathrm{d}x \\ &= \int (1+\log \rho_t)\nabla\cdot(\rho_t v_t)\,\mathrm{d}x \\ &= -\int \rho_t v_t \cdot \nabla \log \rho_t \,\mathrm{d}x \\ &= -\int v_t \cdot \nabla \rho_t \,\mathrm{d}x \\ &= \int \rho_t \nabla \cdot v_t \,\mathrm{d}x.\end{aligned}$$

Interpretation along the flow

The density formula gives $\log \rho_t(x_t) = \log \rho_0(x_0) - \log J_{0,t}(x_0)$. When these terms are integrable,

$$S(\rho_t) - S(\rho_0) = \mathbb{E}_{\rho_0}[\log J_{0,t}].$$

Manuscript §1.2.4, Assessment 9.10. Return to differential entropy.

Appendix A14: the Gibbs variational identity

For bounded U, the laws $\nu_U = Z_U^{-1} e^{-U} \nu$ and ν are equivalent. If $\mu \ll \nu$, the Radon–Nikodym chain rule gives

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu_U} = Z_U e^{U} \frac{\mathrm{d}\mu}{\mathrm{d}\nu}, \qquad \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu_U} = \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} + U + \log Z_U .$$

Integrating under μ yields

$$\mathrm{KL}(\mu \| \nu_U) = \mathrm{KL}(\mu \| \nu) + \int U \, \mathrm{d}\mu + \log Z_U .$$

If $\mu \not\ll \nu$, both relative entropies are infinite. Bounded U keeps the identity meaningful in the extended-real sense.

Minimization

$$\mathrm{KL}(\mu \| \nu) + \int U \, \mathrm{d}\mu \geq - \log Z_U ,$$

with equality precisely when $\mathrm{KL}(\mu \| \nu_U) = 0$, or $\mu = \nu_U$.

Manuscript §1.2.4, Assessment 11.5. Return to Gibbs tilting.